

Search sheets: how to run the activity



The pages after this one are 36 **search sheets**, built from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 54–55 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **in the**, the boxed token is the previous token and *the* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 11 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *the*? it’s a brown one” — and the room can find the swatch before reading a word:

black red brown ochre green teal blue navy violet purple wine

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.

4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

the token pair that answers

in the

on the board

in the

Round 1. Call out the last token on the board (**the**). Another sheet has a match, so that person raises a hand and reads out *milk*.

the milk

in the milk

Round 2. Call out the last token on the board (**milk**). Another sheet has a match, so that person raises a hand and reads out *on*.

milk on

in the milk on

Keep going for as long as you like. The text the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

in the game that fish in said ,
things will Why do what would . that
kites on tell me like , did not
the fish is not can have so ,
a pot fly kites find this I will
the ball tip of sad kind you some
fan ! us this , wet ! so
have to do as should not ! look
wet to had to saw them with the
and I bump on . have I do
he said we do that is milk on
the way . and then I and so
to you ! then not mind , I
will show on a . not

find us

did you

Why ,

not all

! and

and Thing

day .

and the

so tame

, white

said the

hands with

a cake

, what

! said

is a

shame !

wet day

kites ,

I call

. .

. your

with my

then we

can hold

put them

I do

. your

we had

have Lots

. he

tip of

to us

the milk

will not

saw him

you do

the cup

mess ,

on the

fish ,

like to

he put

fan as

do something

house and

! said

that is

the fish

milk and

him !

what to

call up

about it

things and

then , him step mat ! fan ,
put me . you what will house .
? Oh you can call them on a
I said deep and wet , the cat
! and can hold to us . we
and Sally have any ! Thump hop up
with a things I ! Who , I
said the saw him is not the cat
the books mess is , I toy ship
and I I know we saw . and
he said like it hands . milk ,
not a will see . he about with
that are day . shame ! fish !
so so bad tricks do !

what to

him pick

! said

is not

the hat

. Yes

fish .

. two

house !

call this

with a

, with

are pink

said the

want to

toy man

so all

put them

shook their

hands with

you ?

will like

! look

fan with

bad ,

day .

wet and

like these

can hold

? Well

shame !

mess !

I will

! so

on there

he fell

we did

and so

do ?

about .

not know

that is

then Sally

. that

the hat

saw one

to find

and big

things fall

. do

hop on

, said

I will

the cake

deep in

! what

things they

our fish

get them

gown with

fish in

then Sally

. you

wet day

house !

lit .

! and

the toy

toy ship

not sunny

to us

want to

day ?

I can

some good

we looked

he gave

on his

that bump

would you

what a

and Sally

. so

put me

so .

do not

I say

! now

good fun

will show

saw him

the cat

Think of

, said

a ball

can hold

said to

here .

like it

with a

the house

shook up

, no

you should

? Why

and we

are two

. .

there like

. Thing

. then our fish ! and good ,
on a house ? and down . I
Sally . shook with Lots of get out
what the the hat a cup can hold
things yet some fun are good there with
fish as not all the cat he said
so we ? tell toy ship ! I
. But will not . our From up
and Thing to shake would she , and
I do , we put them I wish
you hear like to said the yet !
then I gown ! that . we saw
the hall kind of now you ! bump
here When not be at me

Sally and pick up that is ! and
, how here When so big you sank
their two things in like it hops and
are tame good games not like .
what did , and said the get those
then those fish said ! sit they will
house ! our toy not all Lots of
to be the cat we saw .
? and a cup he came some new
. I on the and Thing ! he
. and I like there , would you
the cat at all gown , now look
bit ! of her can hold I said
and this will she the hat

has gone then , here is things that
and the he was I will a bow
could see there that we had so .
will hit be in at this not like
I will . what . my ! then
can get that is ! we they will
their kites bit ! on the our mother
said the not be ! you some milk
are not fish said pick up and the
? I like that good tricks what to
now ! you wish . I , wet
the fish of the get rid the cat
house ! to get , too bite you
. we Sally and the cake

tail I that is good trick he lit
gave them things . like to dish !
of this be here cold to can fan
! then and the PLOP ! now !
what you on my not . , not
here to they play we saw . .
? what at all ! said then our
not shine house ! a pat hear !
our mother to fall there on you up
could , the rake the cat said the
fish . pick it I can I will
. he so we . But , said
. and Sally and will have For the
bit . ! those the fan

, out

he got

to do

into a

hear ?

! the

a look

, then

! and

and I

our fish

fish shook

. what

were down

pick up

dish ,

things .

I will

not in

could do

Sally and

there is

the hat

be here

here .

at me

on the

the fish

they like

? should

in the

will show

how do

fast as

now !

house all

. she

that were

of this

. .

cold ,

I do

not fly

the hook

said ,

pack up

can get

! Oh

like this

what will

good game

. .

we can

then he

you pack

we saw

a rake

fast as

of good

in the

can hop

will not

. he

be here

kicks and

not be

books !

up and

! I

on the

man !

and all

, she

things ran

I can

could play

our fish

kinds of

give you

, what

that I

cold ,

. we

fish .

you sit

! I

cat go

. and

string of

house For

how that

what would

I hop

the cat

the dots

? Oh

good things

said to

to know

then something

now ,

here When

he shut

. .

at that

Sally and

like the

the house

! look

they like

said to

can have

. these

be here

, we

things they

? now

one and

in with

two things

the day

what a

and the

not want

how I

on the

at me

will give

strings ,

to the

a book

then he

! so

and Thing

! if

box ,

I ,

now look

playthings and

books ,

like it

up these

of something

the pot

mother's new

! they

. then

. and

our house

fish in

. I

we saw

net .

wood box

fast .

, so

you like

that I

lot of

I can

show you

cat in

the ball

they are

good fun

I call show to I can box are
like our one little the house our new
a tip , said ! said to say
you another trick that little toy , the
! her now , net . in the
! put out of , Why . this
cat in lot ! fish said what a
of good . it . and they should
pat . two books . Oh good game
things that said the not be we sat
how . will not the box play .
books ! at all be here and kicks
then he fast as that day and so
? would up the the fish

, with . so . I to the
the things then he trick , fast !
out and we can net , last !
will come looked ! they should the hat
Thing one and a Oh , two things
! do I can fish came pot !
tricks . one and the fish cat in
said the . a in the , do
a box our mother and look now .
at last ! and be about box .
I do you have ! said that is
up on , said show them not one
came down play ! no fear little fish
. this ? Oh of look

no ,

. I

I said

show you

up those

box with

you do

we sat

. have

pot .

will bump

the cake

then our

the mat

two ,

Oh ,

that they

out !

little bit

not wish

Thing two

. .

the ship

and red

looked !

now look

one one

! at

, no

said the

of his

in the

, and

and I

net I

funny !

sat in

, here

they will

cat .

came in

our fish

at me

! those

hall .

! put

sun did

a bump

fish .

play ,

tricks .

to go

. they

I can

? and

of Thing	Oh dear	and the	. there
our fish	box .	in the	. I
little bit	! how	! then	two .
them go	! then	not .	one and
. with	cat in	said ,	play ball
pot .	, what	tricks ,	no fear
Thing two	bow .	and you	hall .
out !	take them	will find	the cat
we sat	this mess	now !	came down
sun is	you get	they want	, the
up up	look at	fish on	I can
if your	book one	at this	the wall
the sun	. that	I let	a lot
, I	to do	sat there	

, jumps

not pick

. you

our game

look !

if you

to do

and the

in a

and a

take a

fall .

When your

of one

, how

do not

we looked

they ran

play .

! said

the hook

the cat

little bit

two things

fear .

. I

the milk

at all

this mess

said the

came down

, look

a big

them a

no fun

I do

out of

! have

home !

I saw

box came

! I

Oh ,

one hand

up high

pot .

run down

sat there

Thing one

his hat

cat .

tricks are

you will

fish to

. it

up . at me ! they cat .

When our in the the cat came Thing

, now of the hand ! the cat

and Thing . . and with Thing

to the , I sat in no fear

you now we tell Oh , the wall

those things I call new tricks play .

do ? they have look at asked you

one ! his hat I always said the

fear ! them out , said . no

out ! fall ! Thing one if I

not know ! then little toy a little

pot . ! said bed ! two things

and I this game

Thing one new . at this a tip
came back this way Well . we saw
the house , and out of not be
. they When your said , cat .
went on and Thing his head , said
and they I had fall ! look at
no fear ! I fear ! ! now
. I . in is what ! he
Make that one ! up all the cat
with a you do of my do .
Oh , tame . was gone thumps and
them out the hat shake hands in the
I can fell on they ran those things
if mother two and

cat with said the out . a fish
! put . said our tame !
those things ! and I bet went away
cake ! I know Make them Oh ,
another good it was they should
new rake up all fear , shake hands
Thing one know how no ! , that
the fish and the fell into . I
wall ! with big not know do not
 . you look at this house and you
the things ! and of his in the
 , said them I one little two !
is not the fish was out mind at
you did say . we could

! they , said sank it Oh ,
my tricks cat , Thing two .
the things was to say . and our
they said two ? the hat . and
the pot , and up the a box
with hops is not fear of sit !
said the of good I stand went bump
look ! bet . ! have ! sit
I did new gown another good
one little and the it , this trick
those two know what mother was not one
you . all the down on do not
them Thing wall in in the cake .
said the . I no fear

one kite

no !

do .

as I

Oh ,

they ran

all .

me !

I sat

my tail

. they

look what

them to

the cat

sank our

not let

. should

fear .

said the

! bump

the cat

come here

, as

and he

is not

, sank

! it

went after

was too

her the

two things

and so

up there

wish we

with a

say .

sit !

. your

cake ,

know !

I like

cat .

of bad

those two

it deep

bet ,

in the

Thing two

said ,

this wet

! not

down the

mother could

a red

was shut

. so

. then

come .

those things

! what

sit !

say ?

step in

! and

is good

up our

cat went

wish to

and some

I know

the string

down with

me .

them away

bumps ,

one kite

I went

, no

a dish

two .

do you

as I

cake on

mother is

look !

with a

to have

, and

! your

the hat

big red

not !

hat .

this box

all kinds

fun to

and our

. now

it a

my playthings

no !

her .

said ,

Thing two

know it

Oh ,

way that

in this

the fish

the cat

to play

cat !

and look

. and

not bad

and he

said ,

in the

away .

it was

a sad

all we

! and

them away

sit there

way !

high as

my hat

her way

mother is

, if

, said

! down

Oh ,

you will

do I

is near

fun in

no way

as fast

I could

was back

Who was

gone with

down my

up up

say !

one and

me now

! said

. you

hat !

two .

this is

big and

with a

Thing one

know some

. I

wish ,

pink ,

look at

picked up	, we	to us	, wet
mother !	is wet	. look	as a
you take	gone away	no !	up !
! look	her about	look at	this !
them run	two and	the house	and the
fun on	a house	I did	. Oh
down !	hat you	Oh ,	away !
cup on	know !	! said	! Make
the cat	ship ,	Thing two	me !
said .	my net	But that	these things
wish to	way home	it came	all !
and he	. with	nothing at	in and
cat .	sit !	not bite	do you
big bumps	with a		

mother will ! fast a game all .
, and up with tall , stand on
fun at ! said hat . know what
do not in the way at them !
big thumps said the not like no .
two hands cat in and the the hall
I will dear . . Oh . But
me down her bed the gown after my
is fun my net these two cup and
with a stop . . a it is
Oh dear this ? you hear , no
But you and the away . look !
Yes . down the to play ! But
ship ! as I

! look it ! I like down !
do was is that dear ! fun ?
! no away ! and then ! have
that cat something went my net
a good , said . . hat .
ball ! her gown ship , said the
But your look ! top of up the
big mess you shook these things as you
mother asked to say all !
have to . . not one the top
cat . me down know some ! we
let you should we with the cup ,
the cat and I , did back in
in a rake . this is no !

this house should not a game fun But
it is , said have no ship and
my net hat . ! and me !
no . down . let down cat in
! said the fish she will ! I
know what look at all ! and I
. and away with ! look you bent
back in up two and we as I
these books in the not wish cup and
hit ! I did But I . then
said the is not mother will do you
the cat top with , we rake ,
with a ball . to stop . I
that they something to

in the ! and all . ! then
ball . ran up to play a shame
my hat and the hat to , as
! and is so did not rake !
kite bump this , you . she do
hook . look at that is down ,
! we up the fun that and Sally
your mother me now should not sunny .
cat . with the as he no !
said the the things have no I can
! . it to But that do ?
something new ! too go !
! they not all the sun any fun
mother came away . , with

so ,

the cat

go away

. now

something to

should not

do !

. take

and she

, Oh

hold up

the box

in on

she say

hat ,

kite we

cat in

ball .

up the

Thump !

have fun

But our

hook .

! I

! sit

with my

I saw

got up

it .

mother is

bump !

ran down

a PLOP

said the

is on

did nothing

to fly

down with

. .

! look

! I

fun ,

look .

not .

your mother

, and

that I

you do

this house

as fast

he should

too .

all the

and a

! But

, little

have no

! and

bump made

. it

and I

is out

fun .

in the

to you

do not

all .

he picked

! on

with a

should not

that !

so deep

did you

I know

too wet

it ?

we can

. they

she will

cat .

! then

! put

mother is

your mother

games we

you do

not like

and a

hook .

look at

a little

something !

ran to

Thump !

the things

always pick

head !

, and

go away

said the

ball !

down !

But that

this big

hat !

the cat

hold you

and then

, cold

not like

! no

things had

said the

it up

you fall

will hold

all I

jump !

cat in

ran about

he should

game that

the pot

on top

near !

to sit

with Sally

! Oh

hold up

should we

she say

! you

bump !

your mother

. Oh

did not

go out

that is

mother is

and with

. he

. he

hat .

we shook

shut with

. now

so so

too cold

head of

in the

, and

then we

! and

have Lots

do to

us two

I said

But we

a shame

is no

the ball

dots that things bump hold a and ,
will show us jump cat in to have
then he she said too wet saw all
game fun on her your mother do !
said the the cat ! they have no
a fox ! tell bump From you have
should we with my , fast . you
we two red . hat . he should
, no is no . you . they
I bet . But go ! ! But
and the the fish in the jumps and
that ? shut up But our did not
not like the things so we ran out
all that ! with

will pick

should not

your mother us ?

tell her

do you

I do

to shake

bump !

on the

with fear

you another

then !

the fan

in this

that cold

game .

red wood

! Make

, I

fox ,

fish in

and too

cat .

have fun

not be

had something

a hook

hold up

said the

, Think

rid of

is not

we did

! Oh

made us

. that

! said

see this

. I

he fell

so tall

! Thump

saw those

the fish

things should

can not

hat !

. then

all if

did !

the cat

and then

. Why

not go

white and

the hat

that cat

your mother

! and

cat in

I know

fish .

and the

then he

will she

so do

all my

he ran

! they

did not

on this

said ,

to play

tell her

, the

the strings

bent our

red fan

should not

! I

can ,

us fast

. then

. Oh

and I

hold up

the pot

wet to

. then

a shame

hat came

game ,

bump their

you did

had them

do if

have no

. I

saw mother's

with a

things are

see something

we could

is funny

in the

! Oh

, said

so , have come on the . these
, so the head and I I can
. what do not and then hold the
cat said kites in he has ! But
bump ! we tell should not us ,
not all Why , what should saw her
tell that things , ! not to go
fly kites ! now game that fish !
, the did not that went can do
. no then we said , shine .
with Thing the dish like it in the
will not a lot the cup . so
you will is not your mother had to
hat . wet ,